

Backend 1 - Web API Building with ASP .NET Core

Background

- This test is about developing a backend API server for a book management system.

Task 1: Creating Database

- A MySQL database `backend1_webapi_building_mysql` has been setup on `localhost:6700` with user `root` and password `123456`
- A book consists of the following properties

Property	Description	Example Values
Id	long, unique identifier	1, 2, 3 ...
CountryCode	string, origin of the book	US, UK, FR, CN, HK, ...
Category	long, a numbered category	1, 2, 3, ...
PublishDate	date / datetime, date of publish	2022-10-28 , 2024-10-28, ...
Name	string, book name	Any free text

- Create a table named `Books` and its relevant indices named for the above properties.
- Insert 1000 rows of random data to the table for the upcoming task.
- You may make any reasonable assumptions.

Task 2: Creating Web Api Application

- Create a web api application in ASP .NET Core to serve the web frontend for the below needs.
- Include proper error handling, pagination and input validation if applicable.
- Be aware of SQL injection when dealing with input validation.
- You may make any reasonable assumptions.

1. CRUD APIs for Book

- a. Create a Book
- b. Get a Book by Id
- c. Update a Book
- d. Delete a Book by Id

2. A Search API for returning a collection of Book with the following filtering criterias

- a. CountryCode
 - i. Returning Book with specific CountryCode
- b. Category
 - i. Returning Book with specific Category
- c. Range of PublishDate
 - i. Returning Book with PublishDate falling into a range
 - ii. Returning Book with PublishDate not later than
 - iii. Returning Book with PublishDate before
- d. Name
 - i. Returning Book with name containing specific phrase

- Note that the above criterias may or may not present at the same time.
- For example, the web frontend should be able to search for Books with CountryCode US and PublishDate not later than 2022

3. (Optional) An Advanced Search API for returning a collection of Book with the above filtering criterias, and supports complex filtering (combination of AND, OR NOT), such as

- i. Returning Book with CountryCode = HK or US
- ii. Returning Book with Category = 2 and CountryCode either from HK or US
- iii. Returning Book with Name containing "Wiki" except for CountryCode US

Tools

You may use the following tools useful in this test.

- VSCode
- .NET 8.0
- MySQL Workbench
- Python 3

Rules

- You are allowed to use online resources such as Google during the test.
- External help is prohibited.

Hints and Reference

- For connection to database, you may use one of the following library
 - Entity Framework Core with Pomelo.EntityFrameworkCore.MySql as provider
 - <https://learn.microsoft.com/en-us/ef/core/>
 - <https://www.nuget.org/packages/Pomelo.EntityFrameworkCore.MySql/>
 - MySQL connector
 - <https://www.nuget.org/packages/MySQLConnector/>
- Tutorial: Create a web API with ASP.NET Core
 - <https://learn.microsoft.com/en-us/aspnet/core/tutorials/first-web-api?view=aspnetcore-8.0&tabs=visual-studio-code>